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#include <BluetoothSerial.h>

#include <Wire.h>

#include <MPU6050.h>

BluetoothSerial SerialBT;

MPU6050 mpu;

#define FLAME_SENSOR_PIN 34

#define BUZZER_PIN 25

#define ACCEL_THRESHOLD 0.5 // ▲ Lower threshold = more sensitivity

#define ALERT_COOLDOWN 2000 // 2 seconds cooldown

unsigned long lastBreakAlert = 0;

float lastAccel = 0;

void setup() {

  Serial.begin(115200);

  SerialBT.begin("ESP32_Flame_Vibration");

  pinMode(FLAME_SENSOR_PIN, INPUT);

  pinMode(BUZZER_PIN, OUTPUT);

  digitalWrite(BUZZER_PIN, HIGH); // Keep buzzer OFF (active low)

  Wire.begin();

  mpu.initialize();

  if (!mpu.testConnection()) {

    Serial.println("MPU6050 connection failed!");

    while (1);

  }

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    Serial.println("System ready.");
}

void loop() {

    checkFlame();

    checkVibration();

    delay(100);

}

void checkFlame() {

    if (digitalRead(FLAME_SENSOR_PIN) == LOW) { // 🔥 Flame detected

        Serial.println(" 🔥 Flame detected!");

        SerialBT.println("ALERT: Flame detected!");

        activateBuzzer();

        delay(1000); // Debounce

    }

}

void checkVibration() {

    int16_t ax, ay, az;

    mpu.getAcceleration(&ax, &ay, &az);

    float accel = sqrt(ax * ax + ay * ay + az * az) / 16384.0; // in g

    if (abs(accel - lastAccel) > ACCEL_THRESHOLD &&
        millis() - lastBreakAlert > ALERT_COOLDOWN) {

        Serial.println("⚠ Breaking detected!");

        SerialBT.println("ALERT: Breaking detected!");

        activateBuzzer();
    }
}

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lastBreakAlert = millis();

}

lastAccel = accel;

}

void activateBuzzer() {

digitalWrite(BUZZER_PIN, LOW); // Active LOW: turn ON

delay(500);

digitalWrite(BUZZER_PIN, HIGH); // Turn OFF

}
```